Physics of the Earth and Planetary Interiors, 54 (1989) 340–352
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Magnetic polarity stratigraphy of Pliocene–Pleistocene marine sediments from Rhodos, eastern Mediterranean

Reidar Løvlie [1], Geir Støle [2] and Nils Spjeldnæs [3]

[1] *Institute of Geophysics, Allegt. 70 N-5007 Bergen (Norway)*
[2] *STATOIL, P.O. Box 300, N-4001 Stavanger (Norway)*
[3] *Department of Geology, P.O. Box 1047, Blindern, N-0316 Oslo 3 (Norway)*

(Received January 4, 1988; revision accepted May 9, 1988)

Løvlie, R., Støle, G. and Spjeldnæs, N., 1989. Magnetic polarity stratigraphy of Pliocene–Pleistocene marine sediments from Rhodos, eastern Mediterranean. Phys. Earth Planet. Inter., 54: 340–352.

A magnetic polarity stratigraphy extending from the end of the Gilbert Chron (~ 3.5 Ma) into the Brunhes Chron (0.73 Ma), has been constructed from three partly overlapping sections of an ~ 25-m-thick sequence of marine Pliocene–Pleistocene sediments of the Sgourou Formation, Rhodos. The natural remanent magnetization is carried by detrital maghemite–magnetite phases. Normal polarity magnetic overprints, most likely of viscous origin, were successfully removed from normal polarity zones only. The secondary overprints of the reversed polarity magnetizations cannot be removed by alternating field (a.f.) demagnetization. Polarity zones have nevertheless been unambiguously determined by detailed analysis of the behaviour of the samples during step-wise a.f. demagnetization. The FAD of *Hyalina baltica* (base of Emilian), occurs below the Jaramillo subchron and is the only available biostratigraphic datum. *Cytheropteron testudo* (Pliocene–Pleistocene boundary) is not found in undoubted specimens within this sequence. On the assumption of a continuous accumulation rate, a one-to-one correlation with the established geomagnetic polarity time-scale (GPTS) has been performed. Within this time-frame a thin, but pronounced, volcanic ash layer is dated to ~ 2.95 Ma, coinciding with the age of igneous activity on nearby islands (K–Ar dated to 3.0–3.15 Ma). Age–depth relationships suggest that deposition of the sediment occurred in three distinct phases with different deposition rates, and these are probably related to local tectonic events.

1. Introduction

Palaeomagnetic polarity stratigraphy has proved very successful for high-resolution, relative correlation of non-lithified deep-sea sediments (Opdyke, 1972), and may provide an important chronostratigraphic tool for correlating and dating continental and shallow marine deposits (Johnson et al., 1975). However, while deep-sea sediments accumulate at almost constant rates in low-energy environments, usually retaining excellent palaeomagnetic records of geomagnetic field reversals, shallow marine environments are more variable with respect to depositional conditions, and uplifted sequences appear to be more seriously affected by magnetic overprinting effects (Watkins et al., 1974; Langereis and Zijderveld, 1979). Dating shallow marine sections by correlating a succession of polarity zones with a geomagnetic polarity time-scale (GPTS), may thus be constrained by poor palaeomagnetic data quality and lack of stratigraphic completeness. Still, palaeomagnetic polarity may provide the only chronological framework for elucidating the depositional history in sediments carrying no other reliable chronostratigraphic markers.

The late Cenozoic sedimentary sequences of Rhodos reflect four phases in development related to the general tectonic history of the Aegean region (Meulenkamp, 1985). Previous palaeomagnetic investigations on islands along the outer Hellenic arc have shown that Rhodos (and Crete)

0031-9201/89/$03.50 © 1989 Elsevier Science Publishers B.V.

has not been exposed to significant rotation since Miocene–Pliocene times (Laj et al., 1982). The present geological character of the island was shaped by events commencing in the early Pliocene, characterized by small-scale, differential normal faulting, superposed on a general regional uplift (Meulenkamp et al., 1972). Marine sediments accumulating to the east–north-east of the island during this period were subsequently raised above sea-level by small-scale faulting, and are now exposed as small hills along the north-east coast.

2. Geological setting and sampling

Pliocene–Pleistocene marine clay–silt sediments of the Sgourou Formation are locally exposed in some small near-shore hills at Cape Vagia, on the eastern coast of Rhodos, Fig. 1. A small basin, with ~ 25 m of unlithified sediments, dips gently towards the east. The basement is an impure recrystallized Jurassic limestone. The contact surface is heavily bored by *Lithodomus*, indicating sublittoral conditions. The lower third of the sediments consists of brownish-yellow clays with numerous, more or less, well-preserved fossils, of which bryozoans form the most numerous group. Near the base there are some spectacular concentrations of coarse-branched celleporid bryozoans. In addition there are brachiopods, molluscs and echinoderms (mostly echinids). The macrofossils indicate a depth of 200 m or more, even at the base of the basin, indicating a very rapid transgression. The microfossils include a high percentage of planktonic foraminifera, in support of this interpretation.

The upper two-thirds of the section consists of dark-grey, yellowish-brown weathered clay. Fine-grained laminated intervals, up to 40 cm thick, occur, especially in the lower part. At least one of these consists of an impure diatomite, with pieces of wood and other coarse plant material. The rest of the clay is bioturbated and homogenized with no sedimentary structures, and with only indistinct bedding. The clays are rather fossiliferous, with generally very well-preserved fossils. The macrofossils consist of deep-water bivalves, scaphopods, terebratulid brachiopods and pteropods. The microfossils are dominated by foraminifera, ostracods and coccoliths. Some of the foraminifera are shallow-water forms (epiphytes), evidently transported from shallower depths. This may also be the case with some of the ostracods. The remaining fauna indicate a depth of deposition between 400 and 600 m.

The age of the fauna is definitely late Pliocene to early Pleistocene. The first occurrence of *Hyalina baltica*, marking the base of Emilian, is found at 5.38 m above the ash bed (see below). Because

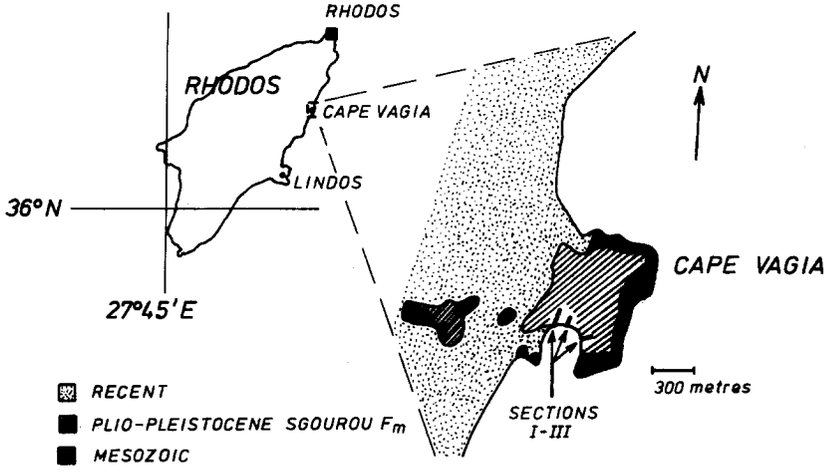


Fig. 1. Map of Rhodos showing the position of Cape Vagia, and a geological sketch map of the sampling locality.

of the wide spacing of samples in this interval, the actual first appearance datum (FAD) may be some 0.9 m lower. *Cytheropteron testudo*, which marks the Pliocene–Pleistocene boundary in the type section, was not definitely identified in specimens in the Cape Vagia section.

In the basal metre of the clay unit, there is a thin (2–4 mm) bright-red layer containing volcanic ash shards. In the finely laminated clays above this ash layer, there are several thin (< 1 mm) streaks of red, interpreted to be redeposited ash or thinner, separate ash falls.

The sequence was sampled along three subsections, laterally separated by ~ 50 and 150 m, respectively (Fig. 1). Elevation measurements were made by hand levelling (± 5 cm) relative to the bright red ash layer which can be followed laterally between sections 1 and 2. Further to the east, a small wadi cuts the sequence at this level, and the positions of samples in section 3 were recorded relative to the most pronounced brownish-red horizon, presumed to be the same ash layer. A total of 176 oriented samples (average sampling interval 28.3 ± 19 cm) were collected by hammering a stainless-steel reinforced brass tube (21 mm i.d.) into the cleaned sediment surface. The sediment entered a cylindrical plastic container (22/23 mm i.d./i.h.) situated inside the brass tube. Each container was closed with a tight lid to reduce further desiccation of the still moist sediment.

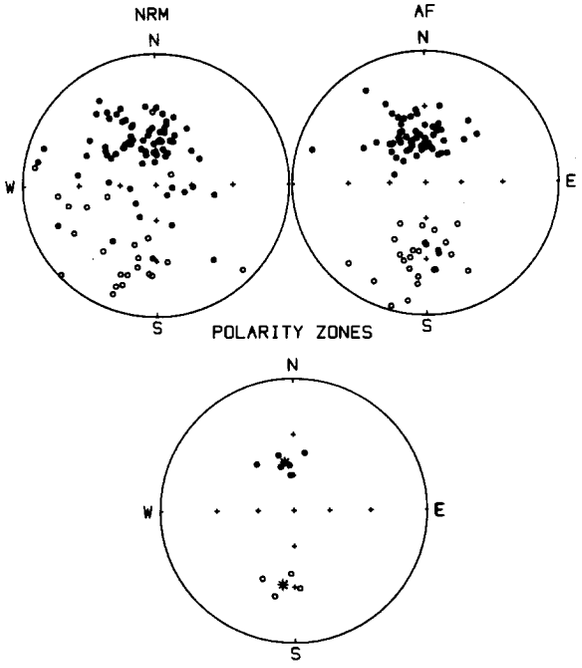


Fig. 2. Stereographic distributions of NRM directions, stable components after a.f. demagnetization and mean directions of corresponding polarity zones. Overall means of the normal/reversed polarity groups are indicated by asterisks. Only reversed zones which incorporate three or more samples are shown (see Table I).

3. Palaeomagnetic properties

The direction and intensity of the natural remanent magnetization (NRM) was measured on a Digico spinner magnetometer. The NRM inclinations define distributions of both polarities, generally confined to opposite hemispheres (Fig. 2). In sections 1 and 2, a more than 10-m-long, consistently normal polarity zone extends from the top to some 8 m above the reference horizon, below which several short normal/reversed polarity zones are present (Fig. 3). Variations in intensity and bulk susceptibility along the stratigraphically overlapping, closely situated sections 1 and 2 are not strongly correlatable, suggesting significant small-scale lateral differences either in depositional conditions, or weathering effects. The reference layer is nevertheless associated with maxima of both parameters. The bulk susceptibility and NRM intensity, ranging from 6 to 190 10^{-5} and 0.4 to 30 mA m^{-1} respectively, exhibit similar variations with stratigraphic height within each section. The close interdependence between these properties is reflected in the significantly smaller amplitude variations in the Q_n' ratio (Q_n' = intensity/bulk susceptibility) (Fig. 3), which decreases systematically with depth owing to an overall decrease in intensity.

Only samples with NRM intensities above 1 mA m^{-1} ($N = 108$) were subjected to progressive alternating field (a.f.) demagnetization. Median destructive fields (MDF) range between 3 and 25 mT, but more than two-thirds of the samples have MDFs confined within the range 12–20 mT. Alternating field demagnetization to 15 mT effec-

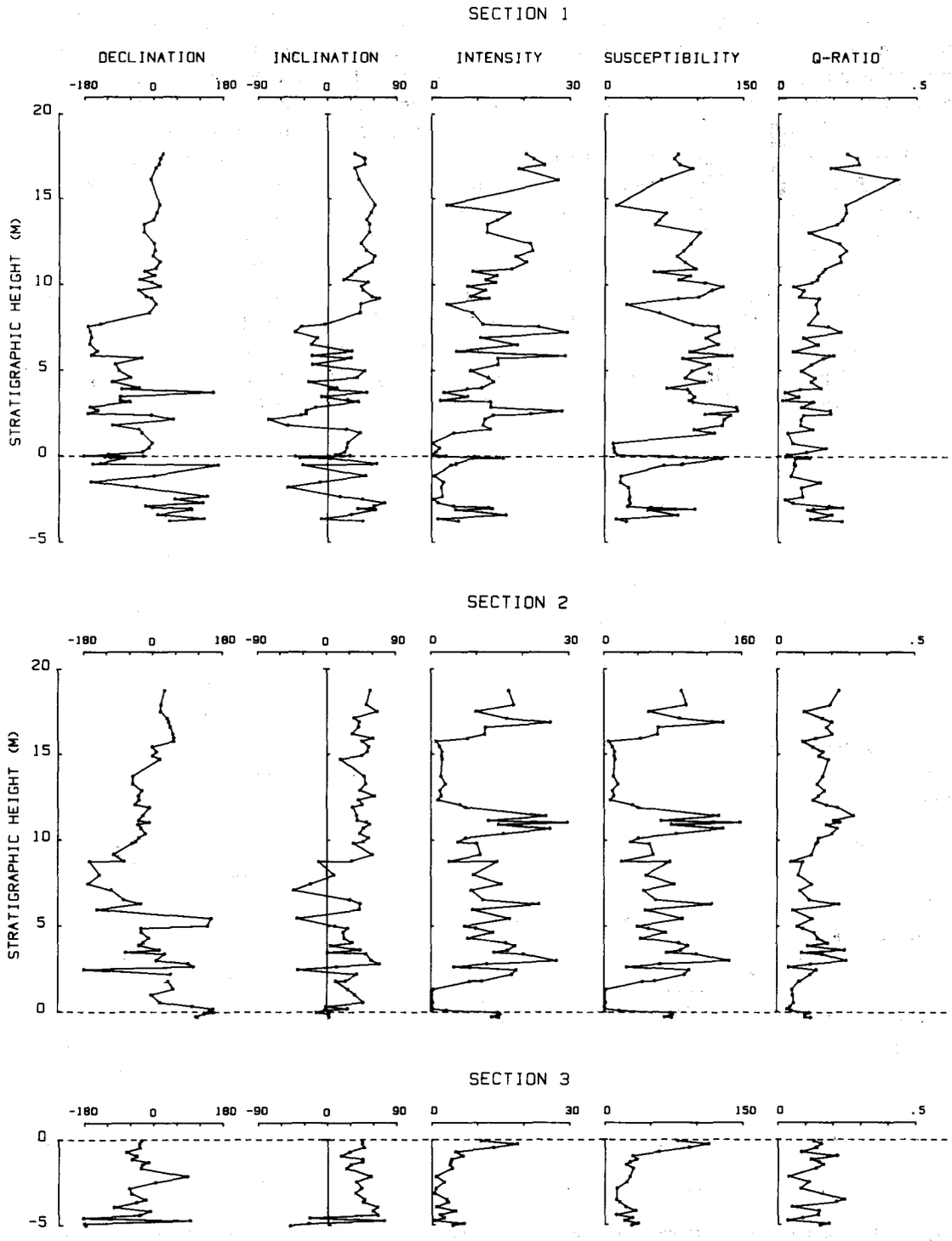


Fig. 3. Stratigraphic variations of NRM direction, intensity (mA m^{-1}), bulk susceptibility ($4\pi \times 10^{-6}$) and Q_n' ratio (intensity/susceptibility) along the three sample sections. Stratigraphic height in metres is relative to the reference horizon which is indicated by a broken, horizontal line.

tively removes highly deviating low-coercivity components. The scattered distribution of these components, defined by three or more consecutive points on linear segments in orthogonal vector projections (e.g. sample V-16, Fig. 4), is evidently not compatible with a magnetic overprint acquired in situ (Fig. 5). Relative to sample coordinates, however, these components define a tight group with inclinations comparable with the ambient geomagnetic field in the laboratory. The samples were all oriented along a common, arbitrary direction during storage, evidently acquiring a significant viscous remanent magnetization (VRM) within weeks. The VRM acquisition of nine demagnetized samples in a 0.1 mT vertical field was monitored between 100 and 1400 h. Viscosity coefficients (S) range between 0.9 and 4.4 mA m^{-1} log t^{-1} (h). Since VRM acquisition appears to be highly dependent on the magnetization history of magnetic grains (Tivey and Johnson, 1981), the palaeomagnetic significance of these high S-values is not considered further. Demagnetization above 3–15 mT results in well-defined, single-component magnetizations of both polarities in 37 samples (34%), (e.g. sample V-52, Fig. 4). A three-component magnetization was recognized in 41 samples (38%) (e.g. sample V-16, Fig. 4). The remaining samples carried unresolvable multicomponent magnetizations reflected by significant great-circle trends, invariably terminating with

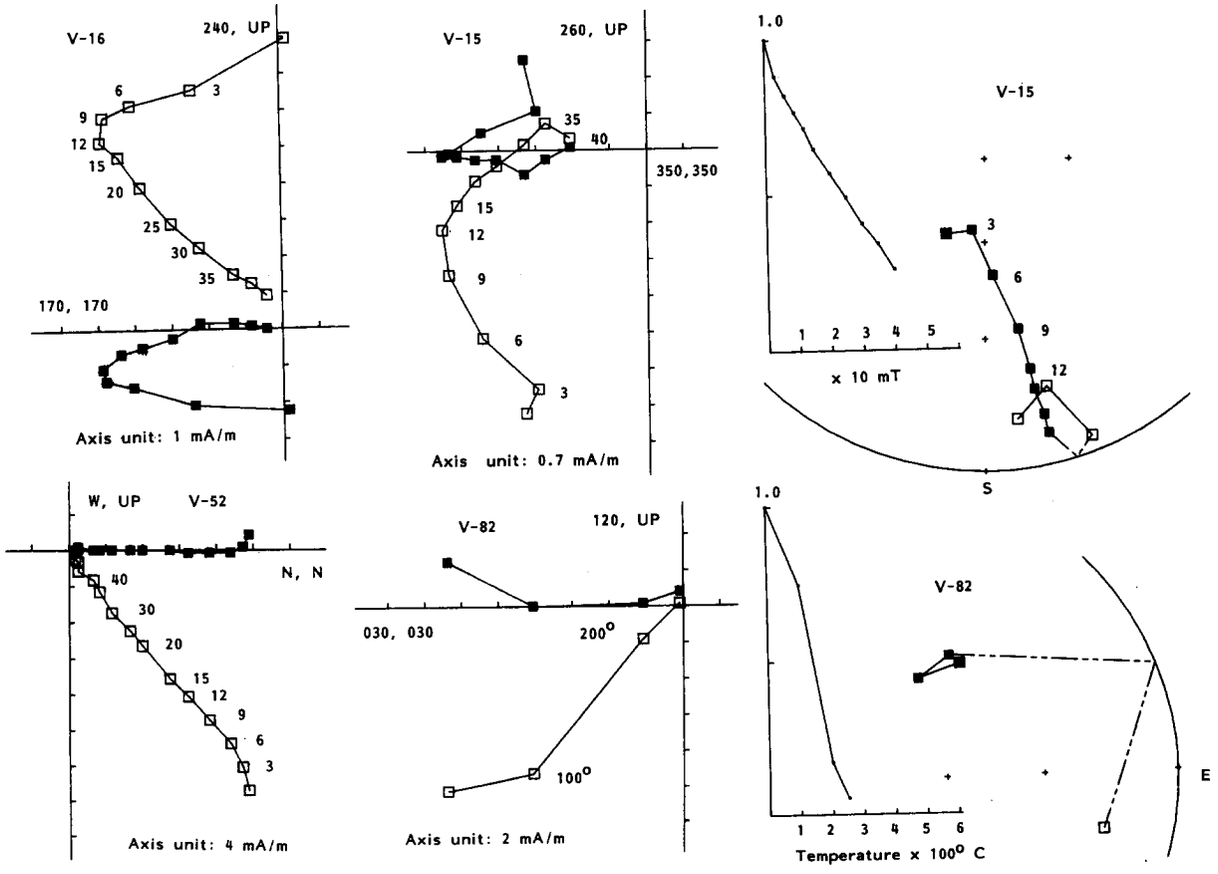


Fig. 4. Demagnetization results in stereographic projection (open and closed symbols represent lower and upper hemispheres, respectively) and optimal orthogonal vector projection. In the latter, the vertical projection plane is rotated along the declination of the characteristic direction (open and closed symbols represent vertical and horizontal projections, respectively). Variation of normalized intensity with demagnetization field and temperature is also shown.

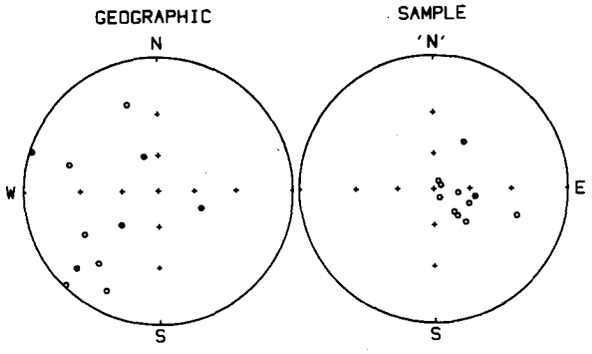


Fig. 5. Distributions of low-coercivity components removed below 15 mT, relative to geographic (left) and sample coordinates (right). 'N' in the latter projection refers to a common but arbitrary fiducial line orientation.

subhorizontal to negative inclinations striking almost due south (e.g. sample V-15, Fig. 4). All these samples are regarded as carrying a reversed polarity magnetization.

The NRM evidently is composed of secondary VRMs acquired in situ and in the laboratory during storage, superimposed on a characteristic, probably depositional-related magnetization.

Stable terminal directions are distributed in two almost antiparallel groups (Fig. 2) representing six normal and six reversed polarity zones (see Fig. 8). While the mean direction of the normal polarity zones is compatible with the present geomagnetic field at the locality, the inclination of the reversed polarity group is significantly shallower (Table I). However, only four of the six reversed zones contain more than three stable directions, seriously affecting the palaeomagnetic significance of the reversed group.

Thermal demagnetization of four specimens, transferred to pre-demagnetized brass containers, revealed characteristic directions associated with pronounced, apparent blocking temperatures between 100 and 250 °C (e.g. sample V-82, Fig. 4). Determination of the NRM remaining above this temperature (< 5%) was seriously affected by transient, viscous components, acquired in the 200 nT vertical field inside the spinner magnetometer.

No systematic directional errors caused by the sampling procedure employed (hammering, shear deformation) have been detected (Symons et al., 1980; Gravenor et al., 1984).

4. Magnetomineralogy

The thermomagnetic properties of the sequence are remarkably uniform. The thermomagnetic

TABLE I

Palaeomagnetic results

Zone	N	Declination	Inclination	α_{95} [a]	k [b]	Mean pole position		
						Longitude (°E)	Latitude (°N)	dp/dm [c]
Normal polarity								
N1	16	355	52	7	22			
N2	3	322	42	17	22			
N3	11	345	43	12	12			
N4	3	355	60	20	15			
N5	7	011	42	17	10			
N6	21	344	52	6	24			
Mean	6	348	50	9	36	274	79	5/7
Reversed polarity								
R1	8	193	−23	10	23			
R4	6	183	−22	15	14			
R5	6	205	−17	24	5			
R6	3	176	−15	10	63			
Mean	4	189	−20	11	25	180	72	20/5

[a] α_{95} = radius of 95% circle of confidence.
[b] k = Precision parameter.
[c] dp/dm = uncertainty in latitude/longitude of palaeopole position.

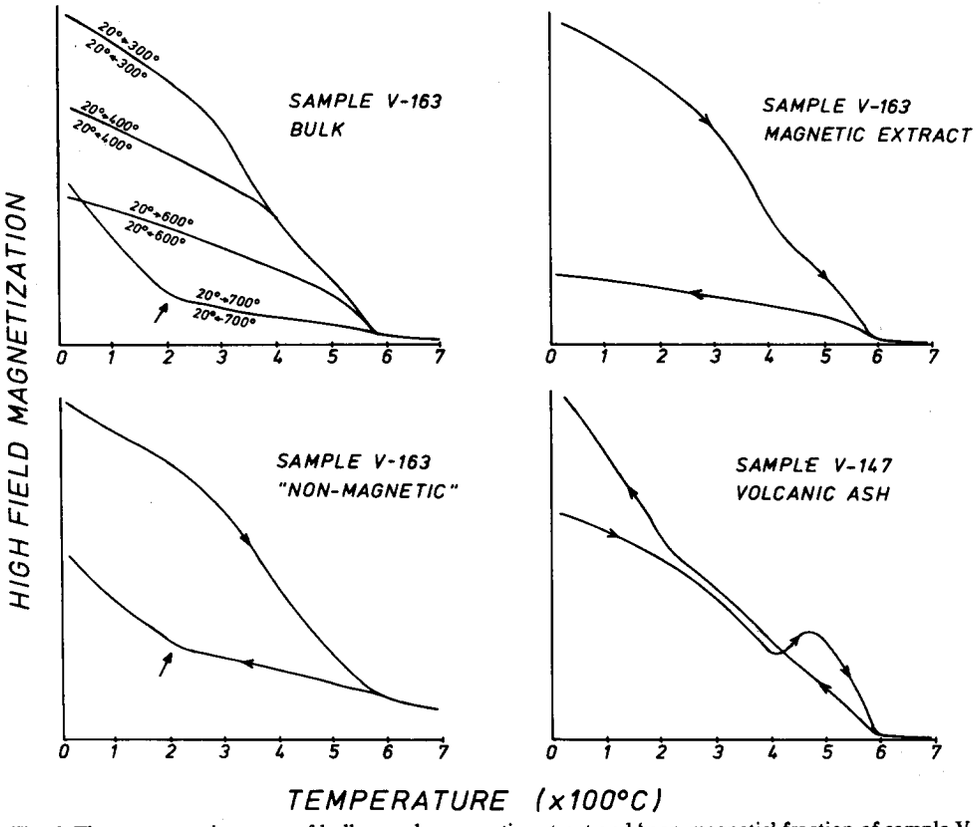


Fig. 6. Thermomagnetic curves of bulk sample, magnetic extract and 'non-magnetic' fraction of sample V-163. The curves suggest the presence of a maghemite–magnetite phase. Note the absence of the low T_c phase after heating the magnetic extract of sample V-163 to 700 °C. The volcanic ash layer contains low-temperature oxidized titanomagnetite. Note the appearance of the low T_c phase after heating to 700 °C.

curves illustrated by sample V-163 in Fig. 6 are typical for the whole sequence, with the exception of the ash layer. Apparent Curie temperatures (T_c) around 300–400 °C are thermomagnetically irreversible, reflecting the inversion of titanomaghemite to haematite. The reversible, high temperature Curie point around 550–570 °C is indicative of almost pure magnetite.

Heating in air to 700 °C creates a strongly magnetic phase with a reversible T_c around 220 °C. Magnetic extraction experiments clearly show that this low T_c phase is created by the breakdown of initially non-magnetic constituents (Fig. 6). The new magnetic phase probably represents a magnetic 'cristobalite', reported to originate from the breakdown of nontronite at temperatures above 900 °C (Moskowitz and Hargraves, 1984). X-ray diffraction of a magnetic extract of this phase revealed both α-quartz and a hexagonal structure identical to that of haematite, in accordance with previous findings (Moskowitz and Hargraves, 1984).

The thermomagnetic curves of the red reference horizon are typical for partly low-temperature oxidized titanomagnetites (Fig. 6), confirming the presence of constituents derived from a basaltic source (volcanic ash shards, H. Furnes, personal communication, 1985). The appearance of the low T_c phase upon cooling even in the ash layer, suggests a rather uniform mineralogical composition of the non-magnetic matrix material.

Progressive isothermal remanent magnetization (IRM) acquisition experiments, reveal complete saturation in fields less than 150 mT, associated

with remanent coercive forces (RCF) between 23 and 40 mT, indicative of magnetite/maghemite phases.

5. Magnetic fabric

The detrital remanent magnetization (DRM) in fine-grained sediments is caused by the preferred alignment of mineral grains with magnetic moments along the ambient magnetic field direction. This alignment may be modified by hydrodynamic forces acting during deposition, or by post-depositional processes (Verosub, 1977). Magnetic fabric is also caused by a spatial orientation of magnetic grains, generally controlled by non-magnetic (hydraulic and gravitational) forces acting during deposition. Thus, magnetic fabric properties reflect the effects of physical processes which may also have affected the acquisition and retention of any DRM. The magnetic fabric is determined by measuring the anisotropy of magnetic susceptibility (AMS), expressed by the length and orientation of three orthogonal principal susceptibility axes ($k_{max} \geq k_{int} \geq k_{min}$) of the susceptibility ellipsoid. Fabric parameters are defined by the following length ratios (Collinson, 1983): lineation = k_{max}/k_{int} (P1), foliation = k_{int}/k_{min} (P3), anisotropy factor = k_{max}/k_{min} (P2) and ellipticity = P3/P1 (E).

The AMS was determined for 45 samples on a commercial induction bridge (KLY-1). Significant anisotropies (P2 > 4%) define a dominantly foliated magnetic fabric (oblate ellipsoids) (Table II), with a tight, vertical distribution of k_{min} axes

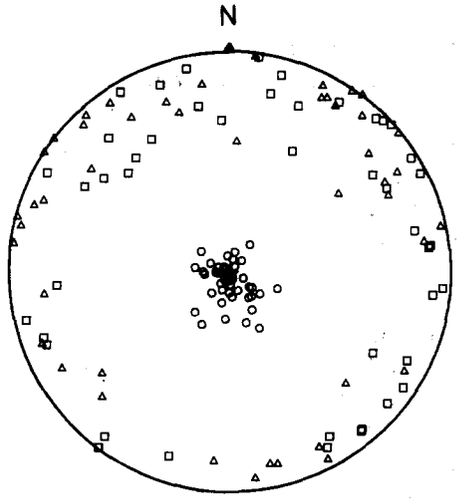


Fig. 7. Stereographic projection of principal susceptibility axes. All symbols are for lower hemisphere: square = k_{max}, triangle = k_{int}, and circle = k_{min}.

associated with subhorizontal k_{int} and k_{max} axes (Fig. 7). These features are characteristic of fine-grained sediments deposited under quiet hydrodynamic conditions on a horizontal surface (Kent and Lowrie, 1975), implying that the inferred depositional remanent magnetization is not likely to have been modified by depositional or post-depositional processes other than compaction.

6. Magnetic polarity zones

The chronostratigraphic significance of palaeomagnetic polarity zones requires that characteris-

TABLE II

Anisotropy of magnetic susceptibility

Section	1	2	3
Number of samples	25	15	5
Mean k_{max} declination/inclination/α_{95}	89/3/16	300/4/20	89/4/35
Mean k_{int} declination/inclination/α_{95}	287/0/15	32/5/20	178/1/35
Mean k_{min} declination/inclination/α_{95}	189/82/4	175/84/6	349/86/10
P1 (lineation)	1.009 ± 0.007	1.010 ± 0.008	1.006 ± 0.002
P2 (anisotropy)	1.056 ± 0.025	1.078 ± 0.033	1.049 ± 0.017
P3 (foliation)	1.047 ± 0.024	1.066 ± 0.034	1.042 ± 0.016
E (ellipticity)	1.037 ± 0.025	1.055 ± 0.037	1.035 ± 0.016

tic remanent directions represent primary, depositional-related magnetizations. The Cape Vagia sequence is characterized by multicomponent magnetizations, which are heavily overprinted by normal polarity directions. Although the latter has been attributed to VRM acquisition during the Brunhes Chron, the presence of titanomaghemite may imply a chemical origin of the overprint as well as of the inferred depositional magnetization. Thermal demagnetization to 250–350 °C, the inversion temperature of the maghemite phase, reduces NRM intensities to less than 5%. Above this temperature, no systematic palaeomagnetic directions appear, suggesting a pre-depositional origin of the maghemite phases.

The assignment of palaeomagnetic polarity zones should preferably rely on directions of characteristic remanent magnetizations defining latitudes of virtual geomagnetic poles (VGP) > ±70 °. This requirement applies to only one-third of the investigated samples. Magnetic polarities of the remaining samples may be inferred from the consistent directional great-circle trends and terminal end-points. This approach is based on the observation that major directional trends towards south/negative inclinations, are invariably due to the removal of normal polarity components, probably of viscous origin acquired during the Brunhes Chron. Hence, high coercivity components are assumed to represent the primary, i.e. depositional-related, magnetization.

The stratigraphic position and polarity of each sample in the three sections, presented in Fig. 8, reveals consistent and comparable patterns of polarity zones along sections 1 and 2. The normal polarity zone 2 (N2), present in section 2 only, can be accounted for by the wide sampling intervals; the corresponding level in section 1 is not represented by any samples. The precision of the stratigraphic positioning is confirmed as being evident for the reversed zone 3 (R3) in sections 1 and 2, which is based on one sample only in both sections. Defining a polarity zone by a single sample is generally of dubious value, but the reliability of this zone is strengthened by the excellent stratigraphic correlation between single samples of the same polarity positioned at the same level (±5 cm) in two laterally spaced sections.

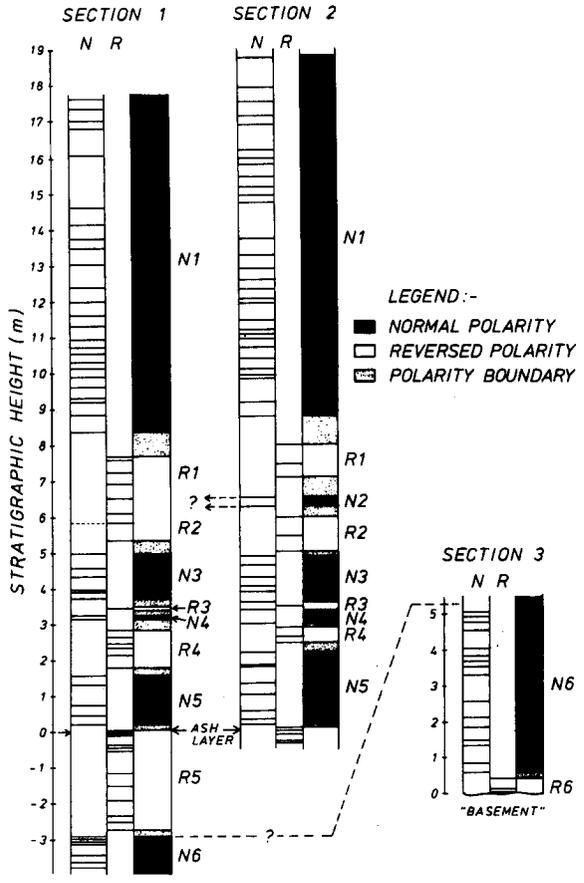


Fig. 8. Stratigraphic positions and palaeomagnetic polarities of each sample along the three sections. The normal polarity zone N2 in section 2 is not present in section 1, but fits in between two successive reversed samples at the appropriate levels in section 1. See text for the stratigraphic position of section 3.

Although polarity boundaries are poorly defined owing to inadequate sampling intervals, the stratigraphic correlation between the polarity boundaries, defined by the distance between two succeeding samples of opposite polarities, nevertheless show a linear relationship (Fig. 9). The only discrepancy is the N5/R4 boundary which does not match owing to significantly different lengths of the corresponding N5 and R4 zones in sections 1 and 2. Section 3 was sampled relative to a red-brown ash (?) layer, which could not be shown to correspond to the reference horizon in sections 1 and 2. In the latter sections, the reference horizon is confined within the top of a re-

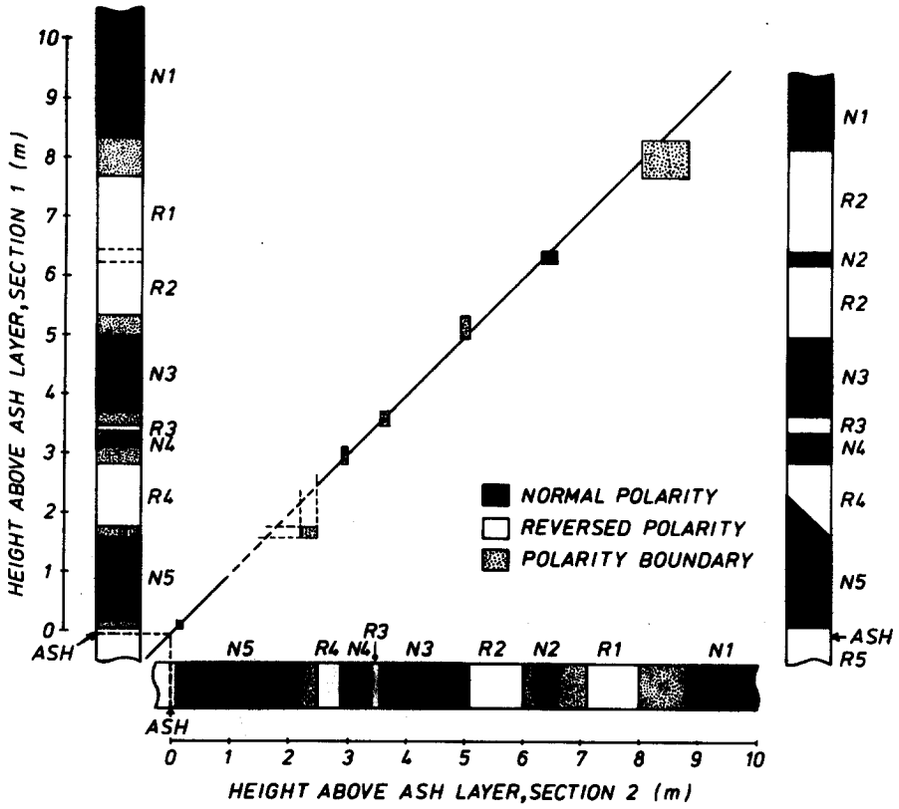


Fig. 9. Stratigraphic correlation of polarity boundaries between sections 1 and 2. Height (m) relative to reference level. The solid line represents a perfect linear relationship. The position of the N5/R4 boundary is uncertain, owing to different lengths of the N5 and R4 zones in the sections 1 and 2. Composite reversal sequence to the right.

versed zone, while in section 3, it is situated at the top of a normal zone. This inconsistency precludes synchroneity of the two laterally separated horizons. While the red ash layer in the former sections is associated with consistent maxima in intensity, susceptibility and the related Q'_n ratio (Fig. 3), the same parameters peak significantly below the inferred synchronous layer in section 3. Values of the Q'_n ratio and RCF are also more in accord with the pattern observed some 4 m below the reference horizon in section 1 at the top of a normal polarity zone (cf. Fig. 3). Hence, the normal zone in section 3 is most likely to correspond to the lowermost normal zone in section 1, which also contains several brown-red layers.

7. Magnetostratigraphy

The composite magnetic polarity sequence at Cape Vagia consists of six normal and six reversed polarity zones. The interpretation of palaeomagnetic polarity sequences should preferably be linked to some reliable chronostratigraphic markers. The FAD of *Hyalinea baltica*, occurring just below the N2 zone (5.35–4.45 m above the reference horizon), is the only biostratigraphic datum level

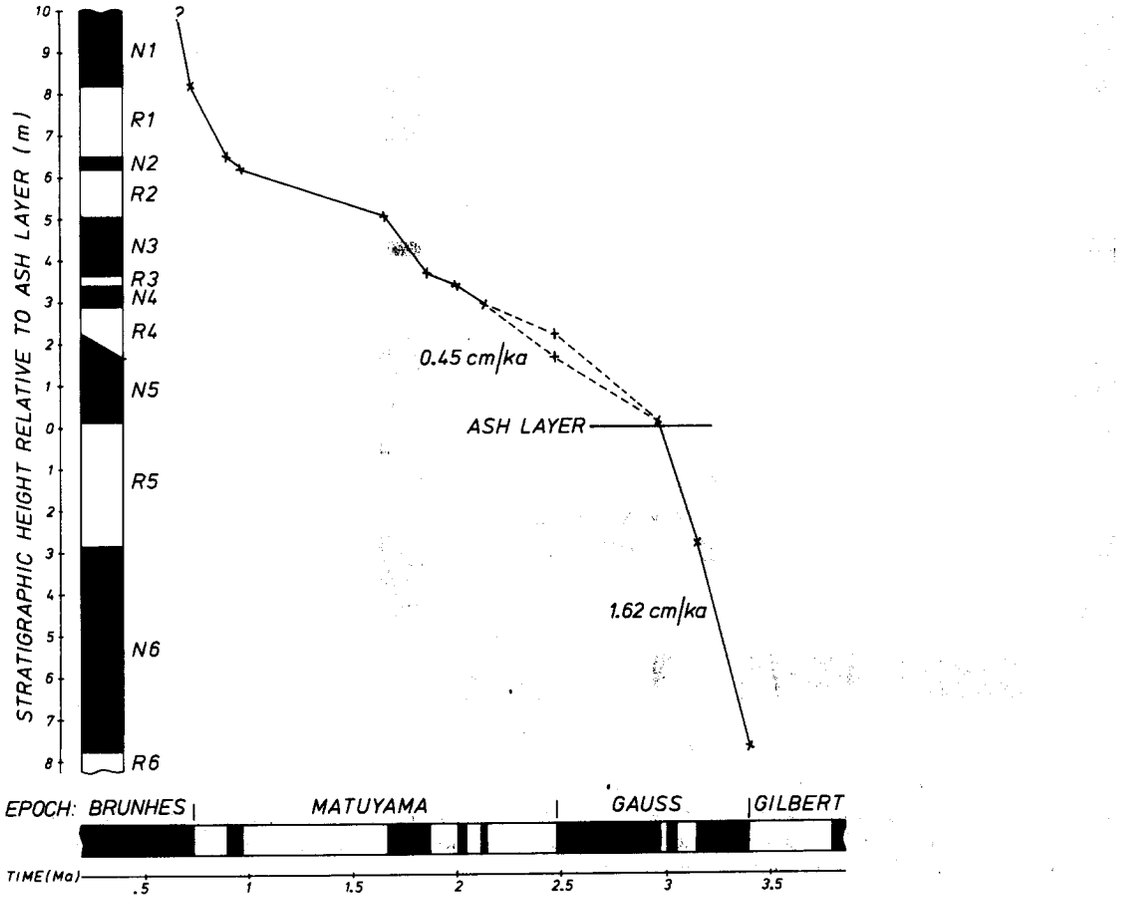


Fig. 10. Age–depth relationship of the Cape Vagia sequence based on correlating the obtained polarity stratigraphy with the magnetopolarity time-scale of Mankinen and Dalrymple (1979). The uncertain determination of the N5/R4 polarity boundary does not change accumulation rates significantly.

in the Cape Vagia sequence. In accordance with findings at the type locality (Aguirre and Pasini, 1985), the N2 zone is interpreted to represent the Jaramillo subchron. Extending the interpretation of the polarity zones must rely on a correlation with a GPTS (Mankinen and Dalrymple, 1979) assuming continuous accumulation of the sequence. This is a reasonable approach considering the lithological uniformity, and results in the age–depth relationship presented in Fig. 10. Owing to the wide sampling intervals, the reversed polarity split Kaena/Mammoth subchron, within the Gauss Chron, appears to be represented by the single normal polarity zone (N5) below the Reunion subchron (N4). According to this interpretation, the composite magnetic polarity sequence at Cape Vagia extends back to the end of the Gilbert reversed chron (represented by the reversed polarity zone R6).

Adopting this chronostratigraphic interpretation, the red ash layer was deposited approximately at the Kaena–Gauss boundary, ~ 2.95 Ma. This age is comparable to ages of igneous rocks on the island of Kimolos and an ash layer on the nearby island of Crete, having K–Ar dates of 3.15 ± 0.15 Ma (Ferrara et al., 1980) and 3.20 $\pm$ 0.03 Ma (Bianchi et al., 1985) respectively, suggesting the onset of a regional magmatic activity

within the southern Aegean region in late Pliocene.

8. Discussion and conclusion

The marine sequence at Cape Vagia accumulated on a horizontal bed acquiring a depositional-related DRM carried by partly maghemitized magnetite grains. The multicomponent remanent magnetization contains VRM overprints of normal polarity which could not be entirely discriminated by a.f. demagnetization. An apparent increase of the viscosity coefficient with depth may either reflect depositional-related variations in composition/grain-size distribution or chemical alterations/dissolution of detrital grains. In general, the results obtained do not give support to the assertion that DRM carried by detrital, partly maghemitized magnetite grains yields significantly higher palaeomagnetic stability than DRM carried by pure magnetite (Kodama, 1982).

The mean pole position for the reversed polarity group samples has significantly lower latitude than that for the normal polarity group. This is attributed to the unsuccessful removal of normal polarity VRM overprints of the former group. Considering that the succession of normal and reversed zones has been exposed to comparable VRM acquisition during the Brunhes Chron, the normal polarity group should also be dominated by an unresolvable normal component. However, the normal polarity zones define a latitude of the mean pole which deviates significantly from 90° N (Table I). Although effects of inclination errors or slight post-depositional tilting of the sequence (Meulenkamp et al., 1972) cannot be ruled out, the results suggest that viscous overprints acquired during the Brunhes Chron have been successfully removed from normal polarity zones. This observation suggests that reversed polarity zones may selectively acquire a relatively higher proportion of high stability VRM compared to normal polarity zones. This may be caused by magnetostatic grain interaction effects, similar to that proposed by Dankers on the determination of the remanent coercive force (Dankers, 1981).

Non-uniform bioturbation may both destroy polarity zones and create false polarity zones, invalidating a direct chronostratigraphic correlation with a GPTS (Watkins, 1968). Polarity zones R5–R2 are situated in laminated sediments, so that effects of biological mixing are unlikely to be important. However, the magnetic fabric properties of the sequence sampled do not indicate any systematic variations between visually laminated and homogeneous sections. The alternation between laminated and non-laminated clays seen in this section, is also found in deep-water sediments in the eastern Mediterranean, where it appears to disturb the fidelity of palaeomagnetic directions.

The magnetic polarity succession has been correlated with a GPTS which does not incorporate two short normal polarity subchrons the Cobb Mountain and Gilsa subchrons which have been reported to occur between the Olduvai and Jaramillo subchrons (Mankinen and Dalrymple, 1979). However, with the accumulation rate inferred from the age–depth relationship (Fig. 10), these subchrons would extend for less than half the average sampling interval.

The FAD of *Hyalinea baltica* in the Cape Vagia section is situated just below the Jaramillo subchron, at variance with the suggested chronostratigraphy at Vrica (Tauxe et al., 1983; Aguirre and Pasini, 1985), in which FAD *Hyalinea baltica* is positioned below a normal subchron between the Jaramillo and Olduvai subchrons. The present results may suggest, therefore, that the N3 subchron at Vrica may in fact be the Jaramillo subchron.

The age–depth curve in Fig. 10 indicated that continuous and fairly fast accumulation (mean rate 1.62 cm ka^{-1}) started around 3.5 Ma. A major drop in the deposition rate (to 0.45 cm ka^{-1}) close to the Kaena subchron (3 Ma) may represent a hiatus of unknown duration seriously affecting the proposed interpretations of the observed polarity sequence. However, we favour the alternative interpretation that the change in deposition rate may be associated with major changes in the depositional environment, probably related to the onset of a regional magmatic/tectonic activity. At the base of the Jaramillo subchron (0.97 Ma), a new phase of fast accumulation commenced, comparable to the initial rate.

Acknowledgements

H. Furnes performed microscopic identification of ash shards. Field expenses were covered by grants from Meltzer High School Funds, and the University of Bergen. Work permit 846/10-2-730, issued by the Institute of Geology and Mineral Explorations, Athens, is greatly appreciated. Constructive comments by E. Hailwood greatly improved the clarity of the manuscript.

References

Aguirre, E. and Pasini, G., 1985. The Pliocene–Pleistocene boundary. Episodes, 8: 116–120.

Bianchi, F.R., Driever, B.W.M., Jonkers, H.A. and Kreuzer, H., 1985. K–Ar date and biostratigraphic position of a volcanic ash layer in the marine Pliocene of Crete, Greece. Geol. Mijnbouw, 64: 103–111.

Collinson, D.W., 1983. Methods in Rock Magnetism and Palaeomagnetism. Chapman and Hall, London, 503 pp.

Dankers, P., 1981. Relationship between median destructive field and remanent coercive force for dispersed natural magnetite, titanomagnetite and haematite. Geophys. J.R. Astron. Soc., 64: 447–461.

Ferrara, G., Fytikas, M., Giuliani, O. and Marinelli, G., 1980. Age of the formation of the Aegean active volcanic arc. In: C. Doumas (Editor), Thera and the Aegean World, Vol. II. Papers and Proceedings of the Second International Congress, Santorini, Greece, August 1978, London, 427 pp.

Gravenor, C.P., Symons, D.T.A. and Coyle, D.A., 1984. Errors in the anisotropy of magnetic susceptibility and magnetic remanence of unconsolidated sediments produced by sampling methods. Geophys. Res. Lett., 9: 836–839.

Johnson, N.M., Opdyke, N.D. and Lindsay, E.H., 1975. Magnetic polarity stratigraphy of Pliocene–Pleistocene terrestrial deposits and vertebrate faunas, San Pedro Valley, Arizona. Geol. Soc. Am. Bull., 86: 5–12.

Kent, D. and Lowrie, W., 1975. On the magnetic susceptibility anisotropy of deep-sea sediment. Earth Planet. Sci. Lett., 28: 1–12.

Kodama, K., 1982. Magnetic effects of maghemitization of Plio–Pleistocene marine sediments, Northern California. J. Geophys. Res., 87: 7113–7125.

Laj, C., Jamet, M., Sorel, D. and Valente, J.P., 1982. First palaeomagnetic results from Miocene–Pliocene series of the Hellenic sedimentary arc. Tectonophysics, 86: 45–67.

Langereis, C.G. and Zijderveld, J.D.A., 1979. Magnetostratigraphy of two Tortonian marine clay sections in western Crete. Ann. Géol. Pays Hell., 1979 (II): 711–716.

Mankinen, E.A. and Dalrymple, G.B., 1979. Revised geomagnetic polarity time scale for the interval 0–5 m.y. B.P. J. Geophys. Res., 84: 615–625.

Meulenkamp, J.E., 1985. Aspects of the Late Cenozoic evolution of the Aegean region. In: D.J. Stanley and F.C. Wezel (Editors), Geological Evolution of the Mediterranean Basin. Springer-Verlag, New York, 589 pp.

Meulenkamp, J.E., de Mulder, E.F.J. and van de Weerd, A., 1972. Sedimentary history and paleogeography of the late Cenozoic of the island of Rhodos. Z. Deutsch. Geol. Ges., 123: 541–553.

Moskowitz, B. and Hargraves, R.B., 1984. Magnetic cristobalite (?): a possible new magnetic phase produced by the thermal decomposition of nontronite. Science, 225: 1152–1154.

Opdyke, N.D., 1972. Paleomagnetism of deep-sea cores. Rev. Geophys. Space Phys., 10: 113–250.

Symons, D.T.A., Stupavsky, M. and Gravenor, M.P., 1980. Remanence resetting by shock-induced thixotropy in the Seminary Till, Scarborough, Ontario, Canada. Geol. Soc. Am. Bull., 91: 593–598.

Tauxe, L., Opdyke, N.D., Pasini, G. and Elmi, C., 1983. Age of the Pliocene–Pleistocene boundary in the Vrica section, southern Italy. Nature, 304: 125–129.

Tivey, M. and Johnson, H.P., 1981. Characterization of viscous remanent magnetization in single-and multi-domain magnetite grains. Geophys. Res. Lett., 8: 217–220.

Verosub, K.L., 1977. Depositional and postdepositional processes in the magnetization of sediments. Rev. Geophys. Space Phys., 15: 129–143.

Watkins, N.D., Kester, D.R. and Kennett, J.P., 1974. Paleomagnetism of the type Pliocene/Pleistocene boundary section at Santa Maria de Catanzaro, Italy, and the problem of post-depositional precipitation of magnetic minerals. Earth Planet. Sci. Lett., 24: 113–119.

Watkins, N.D., 1968. Short period geomagnetic polarity events in deep-sea sedimentary cores. Earth Planet. Sci. Lett., 4: 341–349.